

EARTHQUAKE ALERT SYSTEM USING ARDUINO UNO

A Prototype Embedded Systems Project Report

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Date:

27th June, 2026

1. Problem Statement

Earthquakes are among the most destructive natural disasters, capable of causing severe loss of life and damage to property within seconds. Early detection and warning can play a vital role in reducing these risks by giving people a chance to react before or during seismic activity. Unfortunately, conventional earthquake monitoring systems tend to be expensive and depend on complex infrastructure, which puts them out of reach for many small-scale or educational applications.

The aim of this project is to design and build a low-cost prototype earthquake alert system based on the Arduino Uno microcontroller. The system detects vibrations that simulate earthquake activity and responds with immediate visual and audible alerts through an LCD display, an LED indicator, and a buzzer. In doing so, the prototype demonstrates the core concept of earthquake detection and warning using accessible embedded systems technology.

2. Scope of the Solution

The proposed Earthquake Alert System continuously monitors vibration signals through a vibration sensing mechanism, which is represented in this simulation by a push-button sensor. Under normal conditions, the LCD displays the message “Status: Normal”.

When vibration corresponding to earthquake activity is detected, the system responds in the following ways:

- The event is identified through the sensor input.
- An audible alert is generated by the buzzer.
- A visual warning is provided by the LED.
- The LCD displays an earthquake warning message.

Beyond its core function, the system can serve as a learning prototype for a range of applications, including disaster warning systems, industrial monitoring, embedded system projects, smart safety systems, and IoT-based environmental monitoring. Overall, the project demonstrates how sensors, actuators, display modules, and microcontrollers can be integrated into a single real-time monitoring application.

3. Required Components

Hardware Components

Component	Quantity
Arduino Uno R3	1
16×2 LCD Display	1
Piezo Buzzer	1
LED	1

Push Button (Vibration Sensor Emulator)	1
220 Ω Resistor	1
10 k Ω Resistor	1
10 k Ω Potentiometer	1
Connecting Wires	As Required
PCB Design (EasyEDA)	1

Software Components

Software	Purpose
Tinkercad	Circuit Simulation
EasyEDA	Schematic and PCB Design
Arduino IDE	Program Development
GitHub	Project Documentation and Submission

4. Working Principle

The system is built around the Arduino Uno microcontroller, which acts as the brain of the circuit. A vibration event is simulated using a push-button switch connected to one of the Arduino's digital input pins. The system operates in two distinct states:

Normal Condition

- The sensor input remains inactive.
- The LCD displays "Status: Normal".
- The LED remains off.
- The buzzer remains off.

Earthquake Detection Condition

When vibration is detected, the sequence of events is as follows:

- The Arduino reads the sensor signal.
- The LED turns on.
- The buzzer generates an alarm sound.
- The LCD displays an earthquake warning message.

The alert remains active until the vibration signal is removed.

5. Circuit Description

The circuit is made up of the following key modules:

- Arduino Uno – acts as the central controller and processes the sensor input.
- LCD 16×2 Display – shows the system status and warning messages.

- LED Indicator – provides a visual indication when vibration is detected.
- Piezo Buzzer – generates an audible alarm during earthquake detection.
- Push Button Sensor – emulates vibration sensor activity during the simulation.
- Potentiometer – controls the contrast of the LCD display.
- Resistors – the 220 Ω resistor limits the LED current, while the 10 k Ω resistor acts as a pull-down/pull-up configuration for the sensor input.

6. PCB Design

A PCB layout was designed in EasyEDA based on the verified schematic. The PCB includes footprints for the Arduino Uno, the LCD module, the buzzer, the LED, the resistors, the push button, and the potentiometer. Gerber files were generated successfully and are ready for PCB fabrication.

7. Results

The Earthquake Alert System was successfully simulated in Tinkercad and designed in EasyEDA. The key observations were as follows:

- The LCD displayed the normal system status during idle operation.
- The LED and buzzer activated correctly upon vibration detection.
- The appropriate alert messages were displayed on the LCD.
- The PCB layout and Gerber files were generated successfully.

Overall, the system successfully demonstrated earthquake detection and alert generation using an Arduino-based embedded platform.

8. Conclusion

The Earthquake Alert System successfully demonstrates a simple and cost-effective approach to earthquake warning using an Arduino Uno microcontroller. By providing both visual and audible alerts whenever vibration is detected, the system illustrates the core principles of disaster monitoring in an accessible way. While this prototype focuses on the fundamentals, it can serve as a solid foundation for more advanced disaster monitoring solutions. Above all, the project highlights the practical value of embedded systems in safety and monitoring applications.

References

1. Arduino Uno R3 Documentation.
2. Tinkercad Circuits Documentation.
3. EasyEDA PCB Design Documentation.
4. LCD1602 Datasheet.
5. Piezo Buzzer Application Notes.